



Medi+

AI-Powered Evidence-Based Clinical Q&A Platform

Presented at **Jubilant Pharmova** PPT Presentation Round

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Problem Statement

Clinicians today face a growing set of challenges that impact the quality and speed of patient care decisions. The volume of published medical literature is expanding exponentially, making it nearly impossible for any practitioner to stay current across all domains of their specialty.



Huge Medical Literature

Thousands of new research papers published every day across hundreds of medical journals.



Time Constraints

Clinicians have limited time to search, read, and interpret complex medical evidence during patient consultations.



Misinformation Risk

Unverified sources and AI hallucinations can lead to dangerous clinical decisions.



Lack of Explainable AI

Most AI systems provide answers without transparent reasoning or traceable evidence.



Proposed Solution



✧+ INTRODUCING MEDI+

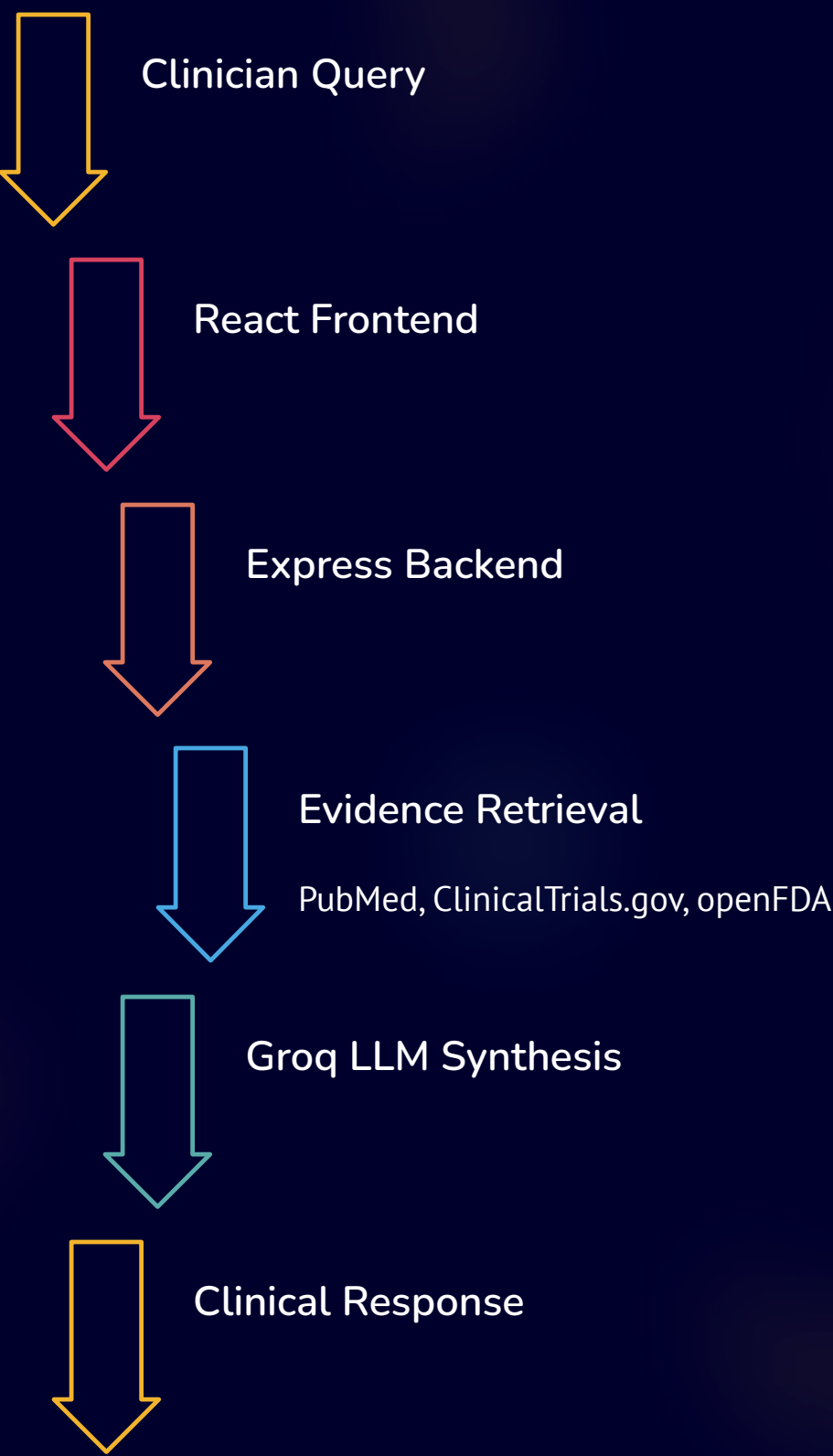
AI-Powered Clinician Support Platform

Medi+ is a purpose-built clinical decision support tool that delivers **trusted, evidence-based medical answers** in seconds. It combines the power of large language models with verified medical databases to give clinicians reliable, cited, and explainable responses.

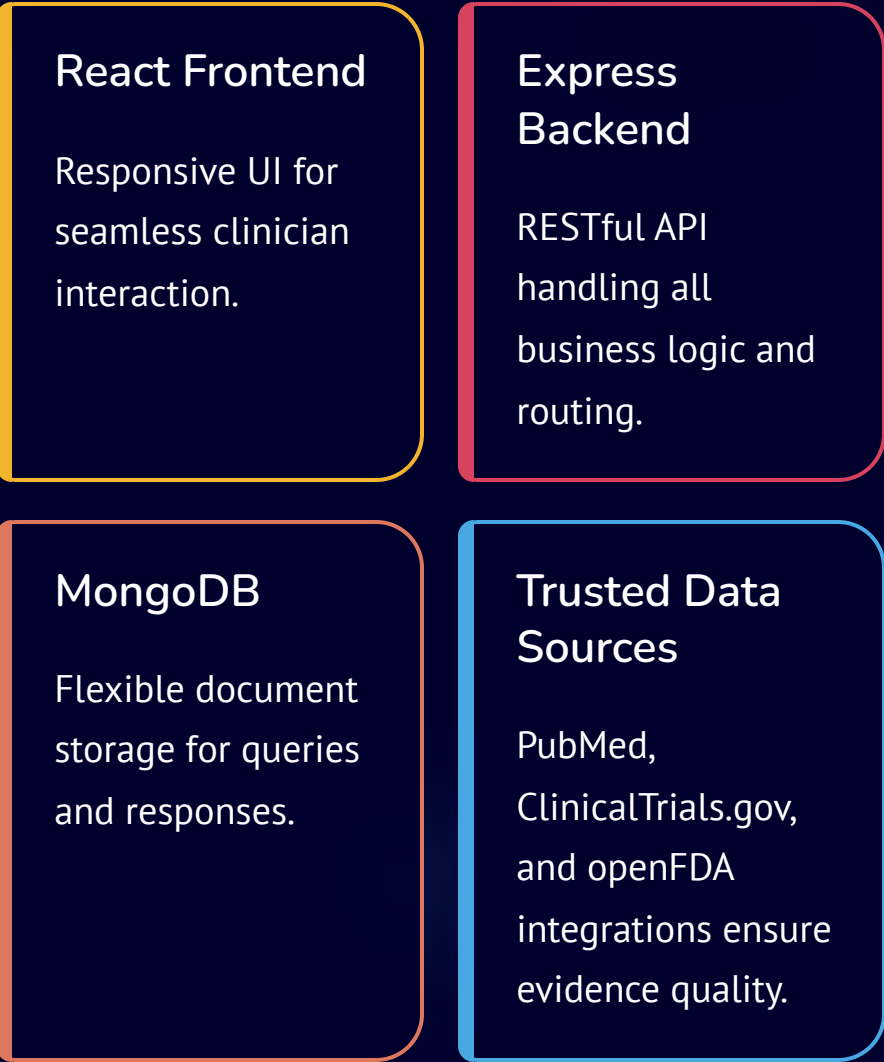
- **Ask Any Clinical Question**
Natural language queries from clinicians.
- **Get Cited Evidence**
Answers backed by PubMed, ClinicalTrials.gov, and openFDA.
- **Make Confident Decisions**
Structured, graded, and risk-analysed responses.

System Architecture

Medi+ is built on a modern, modular architecture that ensures scalability, security, and seamless integration with trusted medical data sources.



Architecture Highlights



Technology Stack

Medi+ leverages a robust and modern technology stack designed for performance, security, and maintainability in a clinical environment.



MERN Stack

MongoDB, Express, React, and Node.js form the core full-stack framework, enabling rapid development with a unified JavaScript ecosystem.



APIs & Integrations

Groq API for ultra-fast LLM inference. PubMed API for literature search. ClinicalTrials.gov and openFDA for regulatory and trial data.



Secure Backend

Modular architecture with role-based access control, input validation, and encrypted data handling to protect sensitive clinical queries.



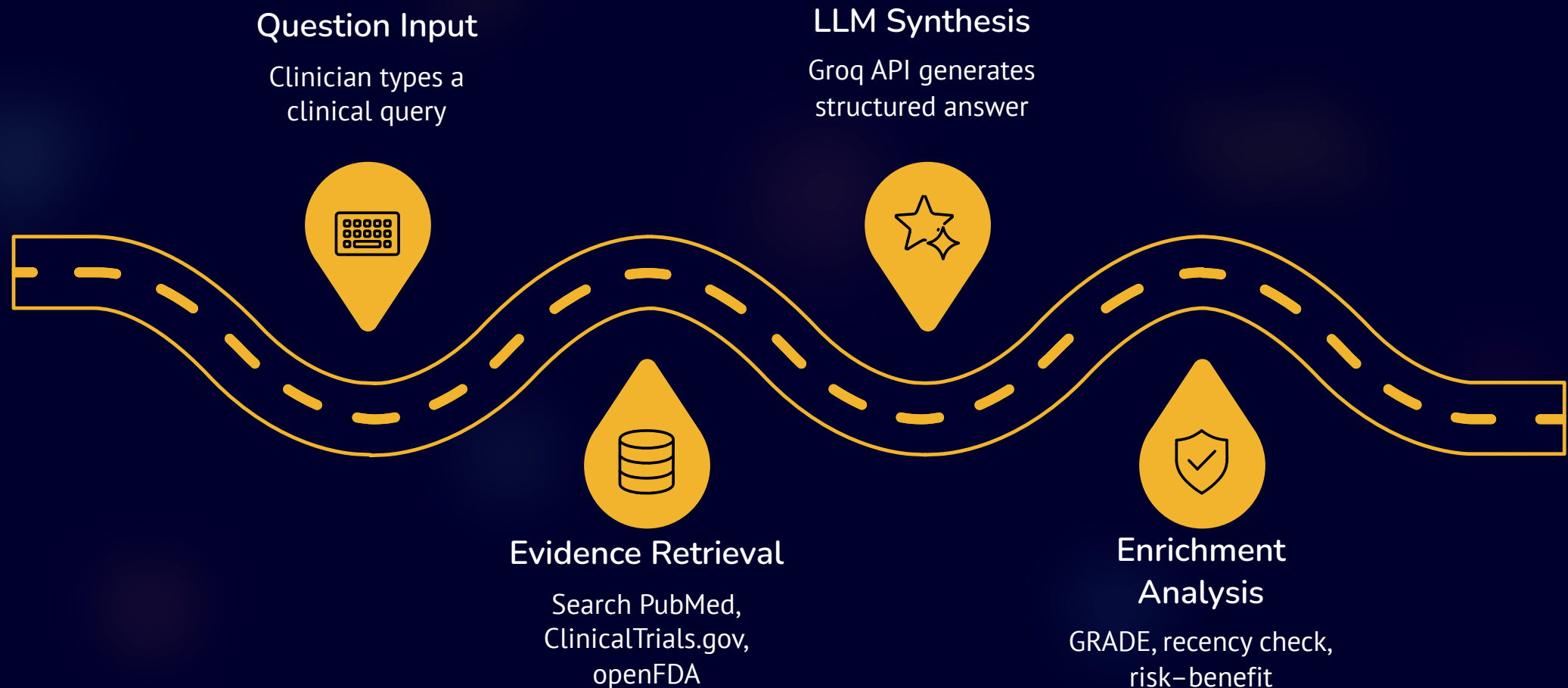
Modular Design

Modular service-oriented architecture enables scalability, maintainability, and easy integration of future AI modules and medical data sources.



Workflow of the System

From question to answer in five intelligent steps – each stage ensures accuracy, relevance, and clinical trustworthiness.



Each step in the pipeline is designed to minimise hallucination risk and maximise clinical reliability, ensuring that every response is grounded in peer-reviewed evidence.

Key Features

Medi+ goes beyond simple Q&A – it delivers a comprehensive clinical intelligence package with every response.



Evidence-Based Answers

Every response is grounded in peer-reviewed medical literature.



Citation Support

Direct references to source publications for verification.



GRADE Scoring

Quality of evidence rated using the GRADE framework.



Recency Alerts

Flags outdated evidence and highlights recent updates.



Risk-Benefit Analysis

Structured comparison of treatment risks and benefits.



Comparative Treatment Analysis

Side-by-side comparison of alternative treatment options.



Unique Selling Points



Trusted Medical Sources Only

Medi+ exclusively queries PubMed, ClinicalTrials.gov, and openFDA – eliminating the risk of unreliable or unverified information entering clinical decisions.



Explainable AI

Unlike black-box models, Medi+ shows its reasoning, cites its sources, and grades its evidence – giving clinicians full transparency into every answer.



Faster Clinical Decision Support

Powered by high-speed LLM inference using Groq API, Medi+ delivers comprehensive, cited responses in seconds – not minutes or hours.

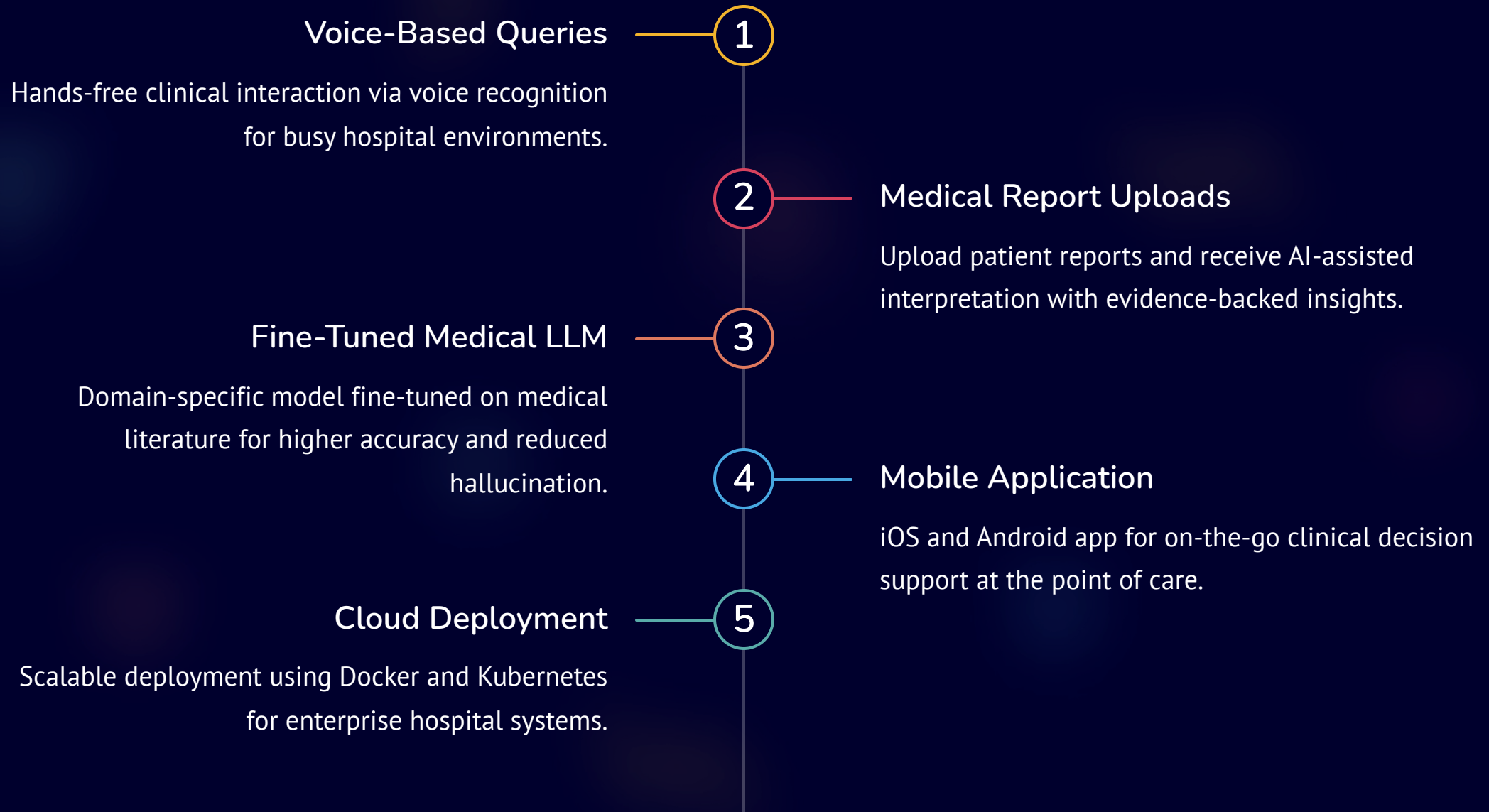


Production-Ready Architecture

Built with enterprise-grade security, modularity, and scalability from day one – ready for real-world clinical deployment.

Future Improvements

Medi+ is designed to evolve. Our roadmap focuses on expanding accessibility, intelligence, and deployment readiness.



Impact & Conclusion

Medi+ has the potential to transform how clinicians access and apply medical evidence in their daily practice.

Improved Patient Safety

Evidence-based, cited answers reduce the risk of harmful clinical decisions.

Reduced Physician Burnout

Faster access to reliable information alleviates cognitive load on clinicians.

Scalable Clinical Decision Support

Production-ready architecture enables deployment across hospitals and healthcare systems.

Medi+ – Empowering clinicians with AI-driven, evidence-backed, and explainable medical intelligence.



Thank You!!

Presented by **Sarthak Morgaonkar**

Jubilant Pharmova